Structural Biology and the Mesoscale: A Technical Research Dossier on David Goodsell

The following report serves as a high-signal repository of raw materials, scenes, and technical data intended to support the creation of a narrative profile for David Goodsell. The analysis prioritizes primary documentation, including peer-reviewed retrospectives, institutional archival records, and extensive interviews. It situates Goodsell within the broader transition of structural biology from manual crystallographic mapping to the computational modeling of the cellular mesoscale.

A) Anchor Scenes

The trajectory of David Goodsell's career is punctuated by moments of technological and conceptual transition. These scenes represent the intersection of physical labor, algorithmic innovation, and the evolution of scientific communication.

The Architecture of the Minor Groove at UCLA

The early 1980s at the University of California, Los Angeles, represented a period where the structural determination of DNA was transitioning from theoretical models to atomic-level empirical evidence. Working in the laboratory of Richard Dickerson, a central figure in the history of crystallography who had previously solved the structure of myoglobin with John Kendrew, David Goodsell was tasked with understanding how small molecules interact with the DNA double helix. The environment was characterized by a specific brand of mechanical rigor that predated the ubiquity of high-performance molecular graphics. Researchers literally lived within the data, surrounding themselves with physical representations of electron density.

Aspect	Details
Date and Place	Circa 1982–1987; Molecular Biology Institute (Boyer Hall), UCLA, Los Angeles.
Primary Actors	David Goodsell (Doctoral Student); Richard Dickerson (Principal Investigator); Irving Geis (Scientific Illustrator).
Methodology	X-ray crystallography of B-DNA oligomers bound to anti-tumor compounds like netropsin.
Technical Texture	Manual stacking of acetate transparencies and plexiglas sheets to visualize electron density maps.

The Turning Point	The first visualization of an electron density map on an Evans and Sutherland MultiPicture System, where Goodsell "saw" individual atoms.

The significance of this scene lies in the realization that scientific methods could transcend mathematical abstraction to provide a literal visual window into the molecular world. Goodsell reflects that this moment was a "big point" in his training—the transition from measuring physical stacks of plastic to interacting with atoms on a vector graphics screen. This experience established the foundational belief that "seeing" is a prerequisite for "understanding" in biochemistry. The collaboration with Dickerson, who was himself deeply engaged in artistic and educational outreach through his work with Irving Geis, provided the cultural permission for Goodsell to pursue visualization as a core scientific endeavor rather than a secondary aesthetic pursuit.

The Volumetric Revolution and AutoDock

By 1990, the challenge of drug discovery had moved into the computational domain, yet the infrastructure remained severely limited. At The Scripps Research Institute, during a postdoctoral appointment with Arthur Olson, Goodsell confronted the problem of "flexible docking"—predicting how a small drug molecule would nestle into the active site of a protein when both could change shape. Most existing methods were rigid, treating molecules like inflexible puzzle pieces. The computational cost of exploring every possible orientation and shape for a flexible molecule was prohibitive for the hardware of the era.

Aspect	Details
Date and Place	1990; The Scripps Research Institute, La Jolla, California.
Primary Actors	David Goodsell; Arthur Olson.
Key Innovation	The development of grid-based energy evaluation (volumetric maps) combined with simulated annealing.
Constraint	Nascent computational infrastructure with limited storage and speed.
Impact	AutoDock became the first and most widely used flexible ligand docking program.

The innovation that Goodsell introduced was the use of "maps"—precalculating the interaction energy of different atom types at every point in a three-dimensional grid surrounding the protein.This allowed the computer to look up the energy of a drug’s position instantly rather than recalculating the physics for every single step of the simulation. This "volumetric approach"

made flexible docking feasible and laid the groundwork for modern computer-aided drug design.The code, largely written in Fortran, represents a moment where algorithmic cleverness overcame hardware deficiency.

Launching the Friendly Doorway to the PDB

In January 2000, the Protein Data Bank (PDB) was a rapidly growing but formidable archive of coordinates that remained largely inaccessible to those without specialized structural training. The archive had reached roughly 13,500 entries, and there was a growing concern that the "sheer magnitude" of the data was scaring off potential users from the broader biology and education communities. Goodsell, now working with the Research Collaboratory for Structural Bioinformatics (RCSB), proposed a monthly column that would tell the "story" of a single molecule.

Aspect	Details
Date and Place	January 2000; RCSB Protein Data Bank.
Primary Actors	David Goodsell; Helen M. Berman.
Aesthetic Choice	A "flat-color and black-outline" style designed to simplify while retaining atomic accuracy.
Initial Molecule	Myoglobin (a direct homage to the roots of structural biology).
Result	Over 300 columns reaching millions of readers and defining biochemistry education for two decades.

This scene marks the institutionalization of Goodsell’s visualization philosophy. By deciding to use a "cartoony" style—inspired in part by the simplicity of artists like Henri Rousseau—he broke the "illusion" of the photograph to provide a "pictorial overview" of life's machinery. This decision was initially met with some skepticism regarding accuracy, but Goodsell argued that artistic freedom was essential for educational clarity. The column became a "friendly doorway," transforming the PDB from a cold data repository into a narrative of biological function.

Real-time Synthesis of the SARS-CoV-2 Virion

The onset of the COVID-19 pandemic in early 2020 presented a unique challenge for structural biologists: how to represent a virus for which data was emerging in disjointed fragments. Goodsell began a series of watercolors that acted as a "real-time integrative tool," attempting to capture the rapidly evolving state of knowledge regarding the virus’s architecture. This was not merely an artistic project but a synthesis of structural biology, microscopy, and biophysics in a moment of global crisis.

Aspect	Details
Date and Place	February – August 2020; The Scripps Research Institute / Home Office.
Scientific Data	Transition from early assumptions of symmetrical spikes to cryo-EM data showing haphazard distribution.
Materials	Watercolor on paper; digital synthesis of PDB and EMDataBank structures.
Milestone	The August 20, 2020, cover of *Nature* depicting the virus surrounded by blood plasma and neutralizing antibodies.

The narrative power of this scene comes from the "deluge of information". Goodsell was often planning the next iteration of his painting before the current one had dried. When early cryo-EM scans revealed that the spike proteins were fewer and more randomly scattered than the "beautiful and uniform crown shape" initially imagined, he immediately adjusted his work to reflect the messy reality of the virus. This work helped the public "gain agency" during a period of fear by making the invisible threat a tangible, physical entity.

B) Direct Quotes From the Scientist

The following selections represent the cognitive framework of Goodsell, emphasizing his focus on visualization as a tool for scientific intuition.

- "Here I am *seeing* these individual atoms... that was a big point in my training. The visual and tangible approach to understanding how things work by seeing their structure became the foundation for his entire career.".
- "When I started, I wanted to create a friendly doorway to the PDB. The PDB contains many interesting structures, but it can be daunting to people who aren't experienced with atomic coordinates and molecular viewers.".
- "I have always hoped that my illustrations spark excitement about the subcellular world, and help us understand the unfamiliar structures and processes that bring our cells to life.".
- "The thing that drives me continually is the beauty of these objects that I'm working on and being amazed at how unusual they are.".
- "Once you draw the outlines, then it's all just paint-by-number?... He was almost correct—in fact, the style is very simple: the colors are painted in, light to dark, and then the outlines are added at the end.".
- "I follow the information as it comes out... I often found myself planning for the next iteration even before the paint had dried on my work-in-progress.".
- "Artistic freedom is essential for a comprehensible and memorable illustration used for educational purposes... while figures appearing in original scientific publications must strictly represent the data.".

C) Human Reality Details

The persona of David Goodsell is defined by a blend of formal scientific rigor and an inherited artistic sensibility. He occupies a dual appointment that reflects this hybridization.

- **Artistic Pedigree**: Goodsell began painting in early childhood under the tutelage of his grandfather, who was an accomplished artist.
- **Educational Choice**: Despite his artistic passion, he did not study art in college, choosing instead to double major in chemistry and biology.
- **Crystallography Roots**: His graduate years coincided with the dawn of computational structural biology, and he learned to write molecular graphics programs out of necessity while at UCLA with Richard Dickerson.
- **The Mentor Model**: He views Dickerson as the archetype of the "gentleman scientist," noting that his own approach to collegiality and the integration of art was modeled after Dickerson's laboratory culture.
- **Signature Magnification**: Most of his cellular landscapes are painted at a consistent $1{,}000{,}000\times$ magnification, a choice that allows students to compare the relative sizes of different biological structures.
- **The "Mesoscale" Focus**: His research specifically targets the "mesoscale"—the scale between individual atoms and the whole cell—which he describes as the "molecular structure of entire cells".
- **Color as Functional Language**: He uses an "intuitive palette" where blues and greens typically represent the cytoplasm and membranes, yellow represents DNA, and magenta represents ribosomes or viral proteins.
- **Speculation vs. Data**: He explicitly distinguishes between his computer renderings, which are strictly data-based, and his paintings, which contain speculation to fill in "gray areas" of scientific knowledge.
- **Custom Tooling**: Most of his iconic "Molecule of the Month" images are created using a computer program he developed himself during his postdoctoral work, originally written in Fortran.
- **Public Outreach Philosophy**: He intentionally uses "off-the-shelf" software like RasMol in his columns to show visitors that they can create their own molecular pictures without needing a PhD.
- **COVID-19 Resilience**: During the pandemic, he released "coloring book" versions of his virus paintings to help provide a sense of agency to the public.
- **Commercial Skepticism**: When approached by Promega to use his art for packaging, he was initially skeptical about how to integrate their specific corporate yellow ("Promega Sol") but eventually found it resembled a "sunrise".
- **Legacy and Retirement**: He retired from the Molecule of the Month series in January 2025 after 300 columns, having hand-picked Janet Iwasa as his successor.
- **Institutional Presence**: His artwork is featured prominently in major textbooks like *Molecular Biology of the Cell* and on the walls of the National Academy of Sciences.

- **Environmental Context**: He often uses watercolor for cellular interiors because the "transparent nature" of the medium allows him to layer molecules and create a sense of depth in a crowded environment.

D) Science Connection (Intro Biochemistry Bridge)

For students in BIOL 3030, the work of David Goodsell provides the most vivid evidence of the **Central Dogma** and **Molecular Crowding**.

Metric	Detail
Best-fit Topic	Protein Structure and Function / The Central Dogma.
Precise Contribution	Developed the computational methods (AutoDock) and visual paradigms that represent the crowded, heterogeneous environment of the cell at atomic resolution.

Teachable Bridge Explanation: Introductory biochemistry often treats the cytoplasm as a "dilute soup" where molecules float freely to find their targets. Goodsell's work demonstrates **Molecular Crowding**, showing that roughly 30% of a cell's volume is packed with proteins and nucleic acids. This crowdedness significantly affects the kinetics of enzyme activity and the stability of protein folding—concepts central to understanding metabolism. His AutoDock program applies these principles to drug discovery by calculating the free energy of binding (ΔG) through grid-based physics, allowing students to see exactly how a drug's "shape" complements its target.
Scientist's Framing/Analogy: Goodsell uses the analogy that a protein in a cell is like a **"grain of rice in a room"**, and the entire cell in a finger joint would be like filling that room with rice. He describes his process as "simulating what we might see if we could enlarge a cell about a million times," making the invisible flow of information from DNA to RNA to Protein a physical reality.

E) Complications, Disputes, or Controversies

Goodsell's career has largely been characterized by collegiality, yet the nature of "scientific illustration" inherently invites debate regarding the boundary between data and art.

Item: The "Speculation" Dilemma

- **What's agreed upon**: Goodsell's paintings are based on the best available data from the PDB, EMDataBank, and biochemical literature.
- **What's disputed**: Because a complete atomic map of a cell does not yet exist, Goodsell must "speculate" about the size, shape, and presence of molecules in the "gray areas".Some scientists argue that filling in these gaps with "artistic license" might inadvertently mislead viewers or introduce errors that go uncorrected for years.

- **Evidence**: Goodsell explicitly addresses this in interviews, noting that he resists using "circles and triangles" for uncertain areas because it "breaks the illusion" of the biological scene.

Item: The Misappropriation by "Intelligent Design"

- **What's agreed upon**: Proponents of Intelligent Design (ID) frequently use Goodsell's "Molecular Machinery" paintings as evidence for the irreducible complexity of the cell.
- **What's disputed**: Goodsell and the mainstream scientific community view these structures as the product of billions of years of evolution. Critics of the ID movement argue that the "simplified representations" Goodsell creates are being used to support "poor scientific epistemology" by making biological processes appear "designed" in a way that the raw, messy data does not.
- **Evidence**: Essays by Stephen C. Meyer and responses from organizations like the National Center for Science Education (NCSE) highlight this ongoing tug-of-war over the interpretation of molecular imagery.

F) "Not on Wikipedia" Gold

Archival research and institutional retrospectives reveal details that provide deeper texture to Goodsell's development.

1. **The Evans and Sutherland MultiPicture System**: Before the "Point and Click" era, Goodsell had to program this high-end vector system manually using custom software to visualize DNA bound to anti-cancer drugs, a task that was as much about coding as it was about biology.
2. **The "Promega Sol" Sunrise**: In his collaboration with the biotech company Promega, Goodsell initially struggled to include their "Promega Sol" yellow until he envisioned it as a "sunrise over a molecular landscape," a rare example of a commercial branding constraint leading to an aesthetic breakthrough.
3. **Djerassi Residency**: He spent a month at the Djerassi Resident Artists Program exploring the "Origin of Life" through speculative paintings, a period of pure artistic exploration that informs his more recent scientific modeling of minimal cells.
4. **Fortran Legacy**: While molecular art is now associated with modern 3D engines, Goodsell's "signature style" software was written in Fortran during his postdoc, a language typically reserved for heavy physics calculations, not "art".
5. **The "Paint-by-Number" Methodology**: Goodsell's admission that his style is "almost paint-by-number"—painting the colors light to dark first and adding ink outlines at the very end—reveals a pragmatic approach designed for the speed needed to handle the "crowdedness" of thousands of molecules.

G) Flagged "Myth-Risk" Details

While Goodsell’s history is well-documented, certain narrative elements carry a risk of being over-dramatized or simplified.

Claim	Source/Location	Why Sourcing is Weak
The "Eureka" Moment at the Terminal	Various popular profiles	The moment where he "saw" atoms is often described as a single flash of insight, but research papers from that era (1982-1985) show a long, laborious period of software debugging and manual measurements.
The Perfect Accuracy of the Nature Cover	Popular media reports	Often framed as a "photograph" of the virus, but Goodsell produced six distinct iterations as the data changed; the "Nature" version was just one snapshot in a rapidly moving field.
The Rousseau Connection	ASBMB Today	His style is frequently compared to Henri Rousseau, but it is unclear if this was a conscious influence from the start or a comparison made by critics that he later adopted.

H) Lens Recommendation

After synthesizing the evidence, the following lenses are evaluated for their narrative potential.

Lens	Score (0-3)	Justification
Eureka	2	The moment of "seeing" the atoms on the E&S screen is a clear turning point, but much of his career is defined by steady, iterative labor rather than a single discovery.
Failure	1	No major, career-ending failure is documented; his challenges were primarily the "nascent state" of technology in the 1990s.
Mentor	3	The Dickerson-Goodsell lineage is exceptionally well-documented and provides a perfect "before and after" regarding the digitization of structural biology.
Controversy	2	The "Intelligent Design" misappropriation is a compelling external conflict, but it does not drive his internal scientific discovery process.

Human Detail	3	The details regarding his grandfather, his lack of formal art training, and his "paint-by-number" honesty are vivid and character-defining.
Last Experiment	3	The January 2025 handoff to Janet Iwasa provides a powerful and concrete closing scene that bridges two generations of visualization.

Primary Recommended Lens: The Mentor (The Lineage of Vision) This lens frames Goodsell not as a lone genius, but as the final evolution of the "gentleman scientist" archetype pioneered by Richard Dickerson. It uses the physical labor of the "Plexiglas stacks" (Scene #1) as the baseline for a story about how humans learned to "see" at the atomic scale. It highlights how Dickerson's permission to integrate art allowed Goodsell to build the "friendly doorway" (Scene #3) that transformed a cold archive into a global classroom.

- **Best Opening Scene**: Scene #1 (The manual stacking of acetate at UCLA).
- **One Quote to Use Early**: "Here I am *seeing* these individual atoms... that was a big point in my training" (Quote #1).
- **Cautions**: Avoid over-dramatizing the "eureka" moment; emphasize the years of manual measurements that preceded it to maintain scientific accuracy.

Backup Lens: The Last Experiment (The Handoff) This lens uses the January 2025 retirement (Scene #5) as a framing device. It allows the writer to "zoom out" from the handoff of the 300th column to look back at the 25-year journey of the Molecule of the Month. It focuses on the legacy of "storytelling" in science and how Goodsell ensured the "doorway" would remain open for the next generation.

- **Best Opening Scene**: Scene #5 (The transition to Janet Iwasa).
- **One Quote to Use Early**: "When I started, I wanted to create a friendly doorway to the PDB" (Quote #2).
- **Cautions**: Ensure the transition between the 2025 scene and the 1980s backstory is fluid and not just a collection of death/retirement facts.

int-www.chem.ucla.edu
Richard E. Dickerson Biochemistry Seminar Fund

chemistry.ucla.edu
Remembering Distinguished Professor Emeritus Richard E. "Dick" Dickerson

technologynetworks.com
Molecular Science and Art With Professor David Goodsell ...

history.amercrystalassn.org
David Goodsell - ACA History - American Crystallographic Association

sites.rutgers.edu
BIOGRAPHICAL SKETCH NAME: David S. Goodsell eRA COMMONS USER NAME (credential, e.g., agency login) - Sites@Rutgers

pmc.ncbi.nlm.nih.gov
The AutoDock suite at 30 - PMC - PubMed Central - NIH

researchgate.net
(PDF) The AutoDock suite at 30 - ResearchGate

pdb101.rcsb.org
PDB Focus: David Goodsell and the Molecule of the Month

forlilab.org
The AutoDock Suite at 30 David S. Goodsell 1,2, Michel Sanner 1, Arthur J. Olson 1, Stefano Forli 1 1. Department of Integrative - The Forli Lab

en.wikipedia.org
David Goodsell - Wikipedia

pdb101.rcsb.org
Molecule of the Month: Twenty Years of Molecules - PDB-101

rcsb.org
Molecule of the Month Newsletter - RCSB PDB

rcsb.org
Insights from 20 Years of the Molecule of the Month - RCSB PDB

rcsb.org
Molecule of the Month: Celebrating 25 Years of Storytelling and Announcing New Beginnings - RCSB PDB

asbmb.org
David Goodsell: The master of mol art - ASBMB

annex.exploratorium.edu
Traits of Life: Exhibits - Exploratorium

issues.org
Cross Section of SARS-CoV-2 Virus - Issues in Science and Technology

americanscientist.org
Painting a Portrait of SARS-CoV-2 | American Scientist

pmc.ncbi.nlm.nih.gov
Science and Culture: The evolving portrait of a virus - PMC

pdb101.rcsb.org
Goodsell Gallery: SARS-CoV-2 and Neutralizing Antibodies - PDB-101

ccsb.scripps.edu
Cellular Landscapes in Watercolor - Center for Computational Structural Biology

promega.com
A Conversation with David Goodsell - Promega Corporation

download.e-bookshelf.de
The Machinery of Life - download

researchgate.net
(PDF) LIFE: a Review/Essay of "THE MACHINERY OF LIFE" by David S. Goodsell (2010) © H. J. Spencer [24Apr.2021] 8,300 words (13 pages) - ResearchGate

pmc.ncbi.nlm.nih.gov
The Machinery of Life - PMC - NIH

crossexamined.org
Intelligent Design Archives - CrossExamined

discovery.org
Intelligent Design and the Origin of Biological Information: A Response to Dennis Venema | Discovery Institute

angelfire.com
The Case for Intelligent Design, in Laypeople's Language - Angelfire

alta3b.com
عصير كتاب: بصمة في الخلية لـ ستيفن ماير Signature in the Cell By Stephen Meyer

exploreevolution.com
Response to the NCSE's Reply to Explore Evolution on Natural Selection